

- 1 Potatoes are underground organs made by the potato plant, *Solanum tuberosum*.

Fig. 1.1 shows a potato plant.

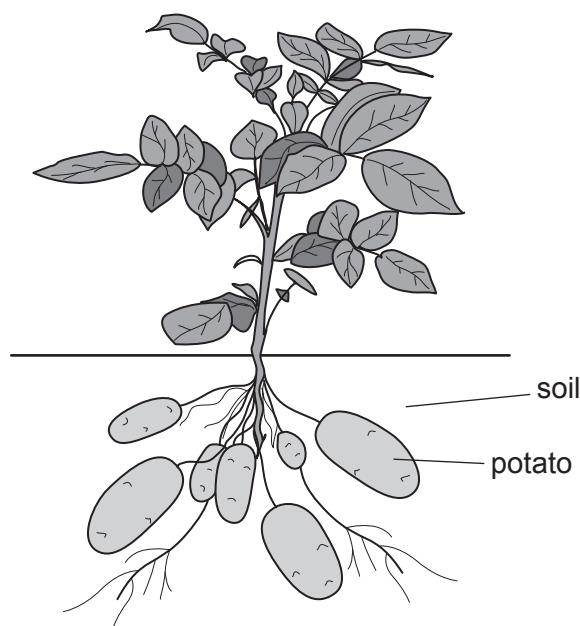


Fig. 1.1

Potatoes contain high quantities of starch and are a popular food. After harvesting, potatoes are stored. During storage, the starch content of potatoes gradually decreases due to the breakdown of starch molecules into glucose.

A glucose assay using a Benedict's test and a colorimeter can be used to determine the concentration of glucose in potatoes.

The glucose assay uses these steps:

- The outer skin is removed from a potato.
- The potato is cut into small pieces that are then ground into a pulp using a mortar and pestle.
- The potato pulp is filtered to remove most of the solids and obtain potato juice.
- Excess Benedict's solution is added to the potato juice.
- The mixture is heated in a water-bath at 90 °C for five minutes. During this five-minute period, copper ions (Cu^{2+}) in the Benedict's solution react with glucose in the potato juice to form an insoluble precipitate.
- The mixture is filtered to remove the precipitate. The filtrate, which is blue, contains copper ions that did not react with the glucose in the potato juice while the mixture was being heated.
- A sample of the blue filtrate is transferred to a colorimeter tube (cuvette).
- A colorimeter is used to measure the absorbance of the blue filtrate.
- A calibration curve is used to determine the glucose concentration of the potato juice from the measurement of absorbance.

Fig. 1.2 shows one type of colorimeter.



Fig. 1.2

A clean colorimeter tube containing the sample to be tested is placed in the colorimeter. Light is passed through the sample and the absorbance of light by the sample is measured.

- (a) (i) Outline how colorimeters should be prepared **before** carrying out measurements on samples so that correct absorbance readings in the glucose assay are obtained.

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To determine the concentration of glucose in potato juice using the glucose assay, a calibration curve needs to be produced. This involves using the glucose assay to obtain absorbance readings for standard glucose solutions of known concentrations.

- (ii) A student was given a 2.0% stock solution of glucose from which to prepare a range of standard glucose solutions of known concentrations.

State the glucose concentrations the student could use to produce a calibration curve **and** describe how **20 cm³** of each solution should be prepared by **proportional** dilution of the 2.0% stock solution of glucose.

You may use a table to show your answer.

glucose concentrations to use:

how to prepare:

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- (iii) Sketch, on Fig. 1.3, a graph of the expected calibration curve that the student would obtain.

Include the axis labels.



Fig. 1.3

[2]

- (b)** The student decided to investigate the effect of storage time on the glucose concentration of potatoes.

- (i) Identify the **dependent** variable in this investigation.

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- (ii) The student was provided with freshly harvested potatoes, standard laboratory apparatus and a colorimeter.

Describe a method, using the glucose assay, that the student could use to investigate the effect of **storage time** on the glucose concentration of potatoes.

Your method should be set out in a logical order and be detailed enough to allow another person to follow it.

Details of how to carry out the glucose assay and how to prepare and use the colorimeter should **not** be included.

[6]

Question 1 continues on page 8.

When potatoes are cooked at high temperatures to make potato chips, glucose and amino acids react together. This reaction is known as the Maillard reaction and is responsible for the orange-brown colour of potato chips.

When glucose reacts with the amino acid asparagine, acrylamide is made.

This reaction is shown in Fig. 1.4.

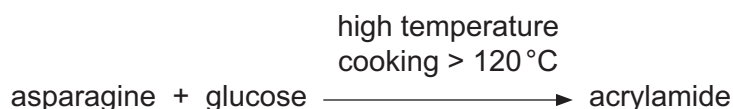


Fig. 1.4

Low concentrations of acrylamide are produced when potatoes are cooked at high temperatures to make potato chips. High concentrations of acrylamide are toxic to humans and can increase the risk of certain cancers developing.

Exposing raw potatoes to gamma radiation decreases the acrylamide concentration of potato chips.

Some scientists investigated the effect of gamma radiation on the composition of raw potatoes.

- Four bags of one variety of potato were each exposed to a different dose of gamma radiation: 0 J kg^{-1} , 50 J kg^{-1} , 100 J kg^{-1} and 150 J kg^{-1} .
- Three samples of raw potatoes from each bag were then analysed chemically to determine the concentrations of glucose, protein and volatile nitrogen compounds as a percentage of dry mass. Volatile nitrogen compounds are present in potatoes due to the breakdown of some proteins and amino acids, such as asparagine.
- Statistical tests were carried out on the data to assess whether the gamma radiation dose affected the concentrations of glucose, protein and volatile nitrogen compounds in the raw potatoes.

Data from the chemical analysis and the statistical tests are shown in Table 1.1.

Table 1.1

gamma radiation dose /J kg ⁻¹	mean composition of the raw potato sample as a percentage of dry mass		
	glucose	protein	volatile nitrogen compounds
0	0.83	10.92	1.14 × 10 ⁻²
50	0.86	11.04	0.75 × 10 ⁻²
100	0.83	10.08	0.68 × 10 ⁻²
150	0.88	10.96	0.40 × 10 ⁻²
statistical significance of differences between raw potato samples exposed to different doses of gamma radiation	not significant at <i>p</i> < 0.05	not significant at <i>p</i> < 0.05	significant at <i>p</i> < 0.05

- (c)** The scientists suggested that gamma radiation reduces the acrylamide content of potato chips by affecting asparagine.

Explain how the information provided **and** the data in Table 1.1 support this view.

..... [2]

(d) In a further study, the scientists investigated whether gamma radiation **and** a hot water treatment reduce the acrylamide concentration of potato chips.

- Four bags of one variety of potato were each exposed to a different dose of gamma radiation: 0 J kg^{-1} , 50 J kg^{-1} , 100 J kg^{-1} or 150 J kg^{-1} .
- After exposure to gamma radiation, the potatoes from each bag were cut into $1 \text{ cm} \times 1 \text{ cm} \times 5 \text{ cm}$ blocks.
- Potato blocks from each bag were given two different treatments.

treatment 1: 100 g of potato blocks were cooked in 1 dm^3 of sunflower oil at 170°C for five minutes.

treatment 2: 100 g of potato blocks were heated in 500 cm^3 water at 85°C for five minutes, removed from the water and then cooked in 1 dm^3 of sunflower oil at 170°C for five minutes.

The scientists measured the acrylamide concentration of the potato chips after these treatments.

This procedure was repeated a further five times to allow statistical tests to be carried out on the results.

The results of the investigation are summarised in Table 1.2.

Table 1.2

gamma radiation dose $/ \text{J kg}^{-1}$	mean acrylamide concentration of potato blocks $/ \mu\text{g kg}^{-1}$	
	treatment 1	treatment 2
0	4551	1768
50	3629	1487
100	3310	1339
150	2073	1010

- (i) Using the data in Table 1.2, calculate the percentage decrease in the mean acrylamide concentration of potato blocks given treatment 2 compared to potato blocks given treatment 1, for a gamma radiation dose of 0 J kg^{-1} .

Show your working.

percentage decrease = [2]

The results in Table 1.2 are shown as a graph in Fig. 1.5. Error bars have been added to show 95% confidence intervals (95% CI).

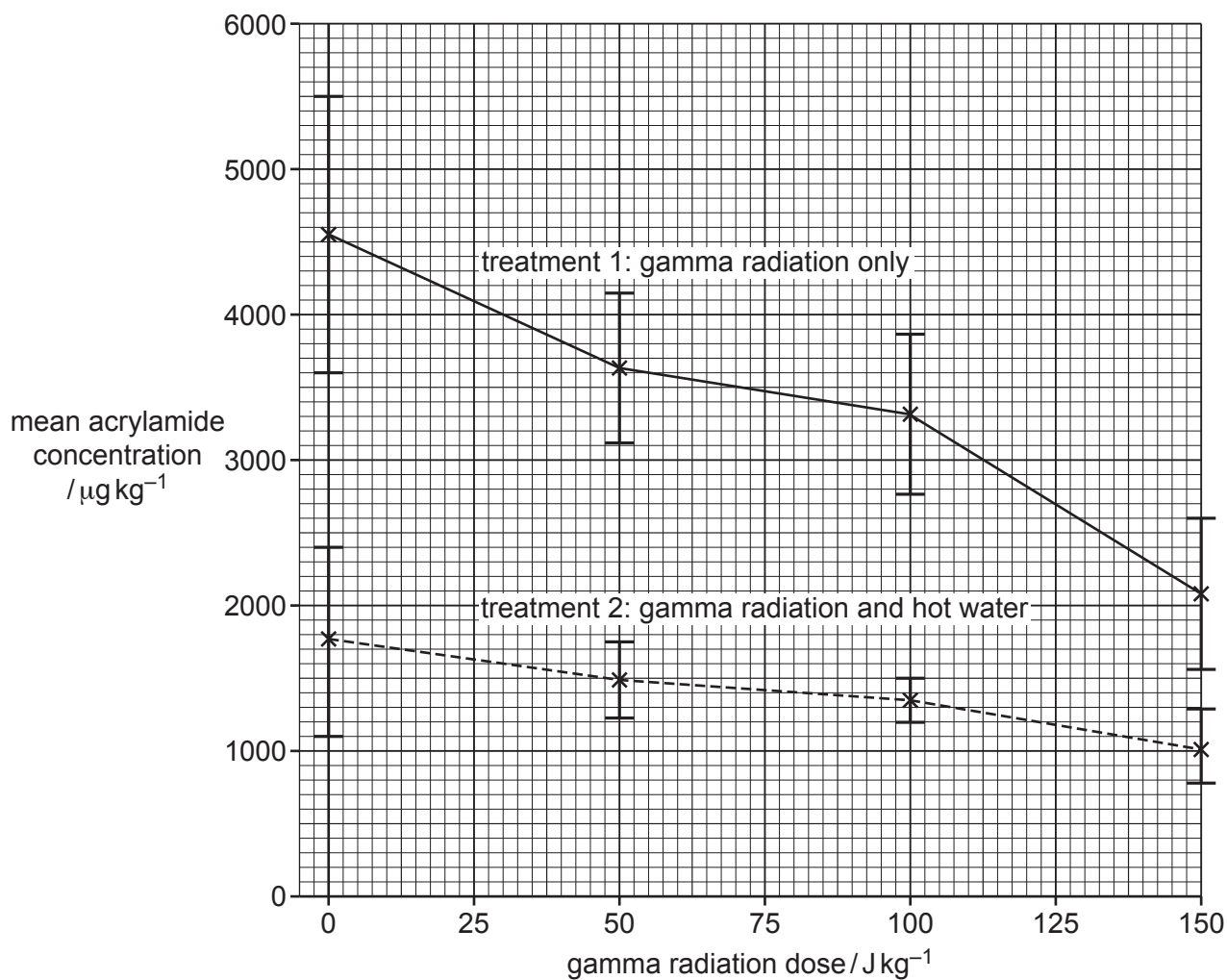


Fig. 1.5

The scientists concluded that a combination of gamma radiation **and** hot water treatment is the most effective way to reduce the acrylamide concentration of potato chips.

- (ii) Explain how the information shown in Table 1.2 and Fig. 1.5 supports, or does **not** support, this conclusion.

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